FRESHROUTE - A SMART FRUIT SUPPLY CHAIN SYSTEM TO REDUCE WASTE AND ELIMINATE MIDDLEMEN IN SRI LANKA

Draft Thesis Report

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The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

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ABSTRACT

Sri Lanka's fruit supply chain is characterized by high post-harvest losses, opaque pricing mechanisms, delayed payments, and a strong dependency on intermediaries who control information, credit, aggregation, and market access. These inefficiencies reduce farmer income, increase consumer prices, and weaken trust between smallholder producers and institutional buyers. This dissertation presents FreshRoute, a blockchain-powered direct marketplace designed to reduce the dependency on middlemen by establishing a transparent, verifiable, and secure digital trading environment for Sri Lankan fruit supply chains. The core contribution of the system is an on-chain reputation framework built around Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs), allowing farmers to develop portable digital trust through successful deliveries, quality verification, and payment histories. The methodology adopts a mixed-methods and design science research approach. Stakeholder requirements are derived through qualitative analysis of farmers, buyers, and intermediaries, while the technical artifact is designed using Hyperledger Fabric, JavaScript-based smart contracts, Node.js APIs, CouchDB state storage, IPFS off-chain evidence management, and React/React Native interfaces. The system includes secure onboarding, farmer supply listing, buyer order placement, intelligent matching and aggregation, smart sales agreement automation, escrow-based settlement, and immutable traceability. Functional, integration, usability, performance, and security testing are planned to validate the prototype. The expected outcome is a practical model for improving farmer bargaining power, reducing payment delays, supporting quality-based pricing, and creating a trusted foundation for future agri-tech services in Sri Lanka. [final pilot metrics and conclusion after evaluation.]

Keywords: Blockchain, Supply Chain, Disintermediation, Smart Contracts, Decentralized Identifiers, Verifiable Credentials, Hyperledger Fabric, Sri Lanka

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My sincere thanks go to my project teammates who contributed to the wider FreshRoute system through related components such as AI demand forecasting, spoilage-reduction logistics, and quality assurance. Their discussions, technical feedback, and collaborative effort helped align this blockchain marketplace component with the broader system architecture.

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
AI	Artificial Intelligence
API	Application Programming Interface
ASMP	Agriculture Sector Modernization Project
B2B	Business-to-Business
CA	Certificate Authority
DEC	Dedicated Economic Center
DID	Decentralized Identifier
DSR	Design Science Research
IPFS	InterPlanetary File System
KPI	Key Performance Indicator
MSP	Membership Service Provider
PHL	Post-Harvest Loss
RBAC	Role-Based Access Control
SaaS	Software as a Service
TLS	Transport Layer Security
UAT	User Acceptance Testing
VC	Verifiable Credential
WBS	Work Breakdown Structure

1 INTRODUCTION

1.1 Background and Literature Survey

Agriculture continues to be one of the most important contributors to rural livelihoods in Sri Lanka. Within this broader sector, the fruit sub-sector has strong potential to contribute to food security, farmer income, local value addition, and export-oriented economic growth. However, the benefits generated by fruit production are not fully captured by smallholder farmers because the existing supply chain remains highly fragmented, intermediary-driven, and vulnerable to post-harvest losses. The FreshRoute research project is positioned within this context and focuses on the design of a digital marketplace that reduces the dependency on traditional middlemen by enabling trusted direct trade between farmers and buyers.

The traditional Sri Lankan fruit supply chain is not merely a movement of goods from producers to consumers. It is an interlinked socio-economic system in which intermediaries provide aggregation, transport coordination, credit, market access, and price information. These functions are valuable in practice, particularly when farmers are geographically dispersed and produce in small volumes. Nevertheless, the same structure also creates information asymmetry and bargaining power imbalance. Farmers often do not have accurate knowledge of real-time market prices, buyer demand, quality requirements, or alternative sales channels. As a result, the farmer's decision-making ability is constrained at the exact moment when produce is most perishable and when rapid sale is necessary.

The literature identifies post-harvest loss as one of the most visible consequences of this inefficient system. HARTI reports have highlighted that fruit losses commonly range from 20 percent to 40 percent, while certain highly perishable fruits such as papaya may experience even higher losses [4]. These losses arise from poor handling, inadequate packaging, lack of cold storage, and the dependence on non-refrigerated road transport. The effect is not limited to wasted produce. The cost of wasted produce is distributed through the supply chain and becomes embedded in the final consumer price, thereby harming both farmers and consumers.

Intermediary dominance intensifies these losses and inequities. Studies on the Sri Lankan vegetable and fruit supply chain show that the cumulative margins captured by various actors can represent a significant portion of the final price paid by the consumer [2]. The structure is reinforced by informal credit relationships in which intermediaries provide pre-season advances or input support. Although this support is useful to farmers with limited access to formal finance, it often creates a lock-in arrangement that compels the farmer to sell to a particular collector at a price determined outside transparent market conditions [9]. Therefore, the central research challenge is not simply the existence of intermediaries but the lack of a trusted alternative that can perform their useful functions without reproducing the same power imbalance.

The national policy environment also demonstrates the need for technological modernization. Initiatives such as the Agriculture Sector Modernization Project aim to improve market linkages, encourage diversification, and increase export competitiveness [5], [11]. At the same time, the Ministry of Agriculture's technology-focused divisions and international partners such as FAO have drawn attention to supply chain weaknesses in transportation, storage, and rural-urban connectivity [8], [10], [13]. These initiatives establish the importance of digital transformation but do not fully solve the trust problem that prevents direct B2B transactions between fragmented smallholders and large buyers.

Blockchain technology has emerged internationally as a potential foundation for supply chain transparency, traceability, and trust. The global implementation of Hyperledger Fabric in food supply chain contexts, such as the Walmart traceability case, demonstrates that permissioned blockchain networks can drastically reduce the time required to trace product origin [15]. However, most high-profile blockchain applications in agri-food supply chains are driven by large corporations with the authority to mandate supplier participation. The Sri Lankan smallholder context requires a different approach: a platform that is accessible to low-resource users, practical for rural adoption, privacy-preserving for commercial data, and capable of creating trust between parties that do not have pre-existing relationships.

FreshRoute addresses this context by proposing a blockchain-powered direct marketplace. Its central innovation is the use of Decentralized Identifiers and Verifiable Credentials to convert a farmer's transaction history into verifiable digital reputation. In this system, each verified farmer receives a DID, and every successful delivery, quality validation, and settlement event can generate or update a credential linked to that identity. This creates a form of portable digital social capital, allowing buyers to evaluate farmers using tamper-evident performance evidence rather than relying solely on informal intermediaries.

Table 1: Post-Harvest Losses in Sri Lankan Fruit and Vegetable Sector

Category	**Post-Harvest Loss (%)**	**Key Affected Produce**	**Primary Causes**	**Source**
Fruits	20 - 40%	Papaya, mango, banana, pineapple	Mechanical damage, rapid ripening, disease, poor handling	[4]
Vegetables	20 - 46%	Okra, tomato, carrot	Unsatisfactory packaging, lack of ventilation, poor handling	[4]
Transportation	19 - 21%	All fruits and vegetables	Improper transport methods and lack of cold chain	[6]

The figures summarized in Table 1 indicate that supply chain inefficiency has both physical and economic dimensions. When a large percentage of produce deteriorates before reaching the final consumer, the loss is not borne by one actor alone. Farmers lose potential income, buyers face supply uncertainty, and consumers pay higher prices for the remaining marketable produce. The system therefore requires mechanisms that encourage quality-based handling, timely collection, better market coordination, and transparent accountability.

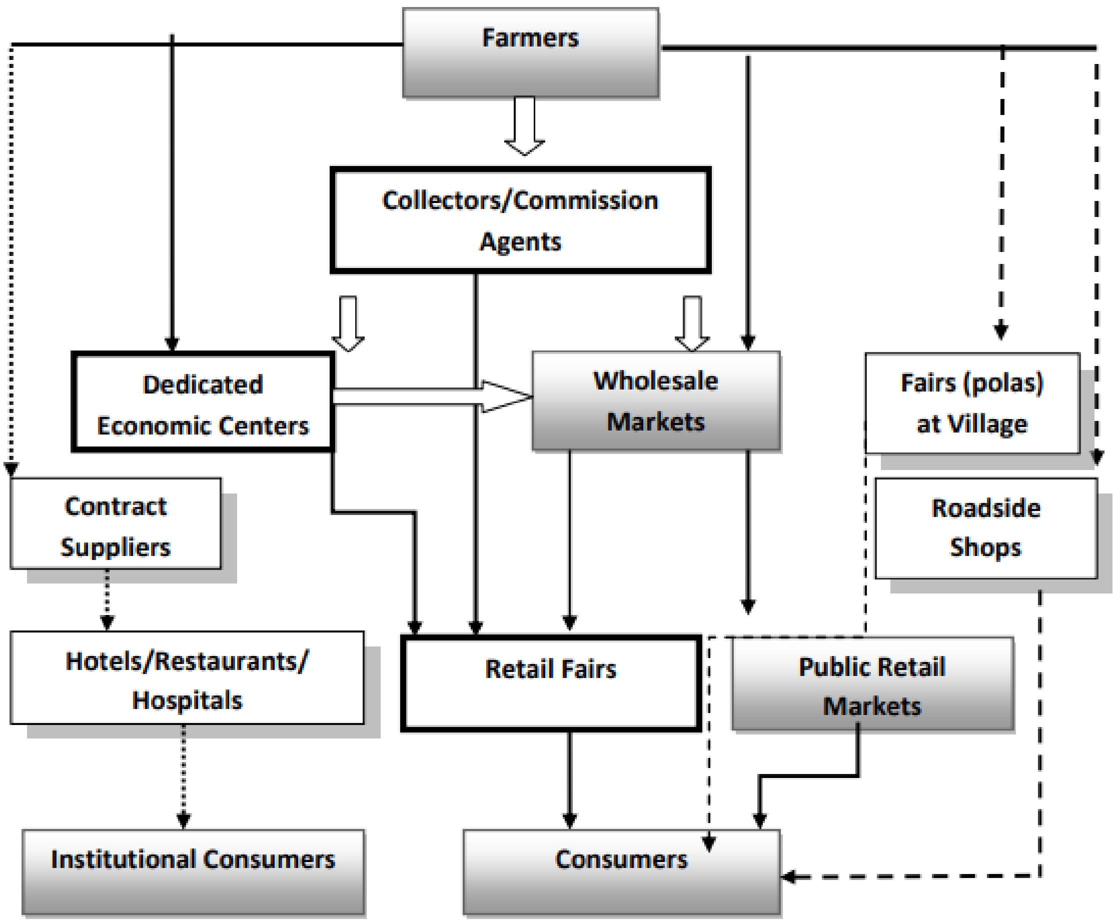

Figure 1: The Traditional Sri Lankan Fruit and Vegetable Supply Chain

Source: Vidanapathirana, Priyadarshana and Rambukwella, 2011

1.1.1 Intermediary control and information asymmetry

The traditional supply chain depends on collectors, commission agents, traders, transporters, and wholesale actors. These intermediaries are not always unnecessary; they provide aggregation and logistics services that individual smallholders cannot easily provide by themselves. However, the problem arises when the intermediary becomes the sole channel through which the farmer accesses markets. In such situations, the intermediary controls price information, determines the buyer relationship, coordinates logistics, and may also function as a creditor. This concentration of functions creates dependency and reduces the farmer's ability to negotiate.

Information asymmetry is particularly harmful because fruit is perishable. Farmers often need to sell quickly after harvest, but buyers may require evidence of quality, quantity, reliability, origin, and delivery capacity. Without a reliable digital reputation system, buyers may prefer to purchase through known intermediaries rather than directly from unknown farmers. Thus, both sides face trust problems. Farmers fear unfair prices and delayed payments, while buyers fear inconsistent quality and delivery failure. FreshRoute aims to transform this trust problem into a verifiable data problem by recording performance evidence on a permissioned blockchain.

Table 2: Cost Factor Contribution to Final Vegetable Price in Sri Lanka

Cost Factor	**Contribution to Final Price (%)**	**Implication**	**Source**
Profit across actors	33 - 44%	A large share of the final price is retained across the chain, with substantial capture by intermediaries.	[2]
Post-harvest loss	9 - 33%	The cost of wasted produce is transferred to the price of remaining produce.	[6]
Commission	5 - 10%	Commission agents charge direct fees at wholesale markets.	[2]
Transport	1 - 3%	Transport cost has increased significantly after the economic crisis.	[3]

1.1.2 Existing digital solutions and remaining limitations

Digitalization in Sri Lankan agriculture has already begun through mobile information systems, smart farming tools, ERP solutions for plantations, and market platforms. Examples discussed in the proposal include GoviLab, Sarvodaya's Fusion ICT4D, SenzAgro, Agrithmics or Cultive8, and Agroworld Web. These solutions demonstrate that technology adoption in agriculture is possible, but they also show that existing interventions are often specialized, localized, or focused on only one segment of the agricultural process.

A direct marketplace that only displays listings would not be sufficient for the FreshRoute problem. If a buyer cannot verify the identity, reliability, and historical performance of a farmer, the buyer will continue to rely on intermediaries. If a farmer cannot verify that payment will be made on time, the farmer will remain reluctant to shift from familiar informal arrangements. Therefore, the missing component is a digital trust infrastructure. FreshRoute contributes to this gap by combining verified onboarding, DID-based identity, on-chain reputation, smart contract escrow, traceability, and intelligent aggregation into a single marketplace framework.

1.1.3 Blockchain, smart contracts, and decentralized identity

A permissioned blockchain is suitable for this research because the target domain is a B2B supply chain with identifiable participants rather than an anonymous public marketplace. Hyperledger Fabric supports this requirement through membership services, channels, private data collections, endorsement policies, and modular consensus. These features allow the platform to provide accountability while protecting commercially sensitive data such as negotiated prices and buyer procurement volumes [17], [20].

Smart contracts, referred to as chaincode in Hyperledger Fabric, execute business logic on the blockchain network. Within FreshRoute, the Smart Sales Agreement governs the lifecycle of a trade. It records the buyer order, matched farmers, agreed quantity, price, expected delivery date, escrow status, quality confirmation, and settlement status. When the required verification event is recorded, the agreement updates the ledger state and releases the payment according to the defined business rules. In a production environment, settlement may integrate with a regulated payment gateway or bank API; in the prototype, the same logic can be simulated using tokenized balances or transaction states.

Decentralized Identifiers and Verifiable Credentials are used to represent identity and reputation. A DID is a globally unique identifier controlled by the subject rather than by a centralized platform. A VC is a tamper-evident claim issued by a trusted party or system process. For example, a completed transaction can result in a credential stating that a farmer delivered 500 kg of grade A mangoes on time, passed quality verification, and received settlement. By linking these credentials to the farmer's DID, the platform creates a reliable historical profile that can be evaluated by future buyers.

Table 3: Research Gap Analysis

Research/Existing Practice	**Sri Lankan smallholder focus**	**Direct B2B marketplace**	**DID/VC reputation**	**Smart contract settlement**	**Intelligent aggregation**
Traditional intermediary system	Yes	No	No	No	Manual
Existing localized agri platforms	Partial	Partial	No	No	Limited
International blockchain supply chain cases	No	Yes	Limited	Partial	Enterprise-led
FreshRoute blockchain marketplace	Yes	Yes	Yes	Yes	Yes

Table 4: Comparison of Blockchain Platforms for Enterprise Supply Chains

Feature	**Hyperledger Fabric**	**Solana**	**Polygon**
Ledger type	Permissioned	Permissionless	Permissionless PoS
Data privacy	High through channels and private data collections	Low due to public ledger visibility	Low due to public ledger visibility

Feature	Hyperledger Fabric	Solana	Polygon
Governance	Consortium-based	Foundation/public ecosystem	Foundation/public ecosystem
Suitability for B2B supply chain	Excellent for known participants, confidentiality, and accountability	Poor for sensitive commercial data	Poor for confidential B2B trading
Source	[20]	[21]	[22]

1.2 Research Gap

The literature establishes three important facts. First, the Sri Lankan agri-food supply chain experiences high losses and inefficient price distribution. Second, intermediaries dominate market access by controlling information, credit, aggregation, and buyer relationships. Third, blockchain technology has been successfully demonstrated internationally as a traceability and transparency mechanism. However, these strands of research do not fully converge in a system that is tailored to Sri Lankan smallholder realities.

The first gap is the absence of a practical blockchain-based direct marketplace designed for the socio-economic conditions of Sri Lankan fruit farmers. Many digital agriculture platforms focus on information dissemination, crop intelligence, investment, or enterprise resource planning. They do not directly address the trust and payment mechanisms that allow farmers and institutional buyers to transact without intermediary dependency.

The second gap is the lack of verifiable, portable farmer reputation. Informal trust is currently embedded in human relationships and intermediary networks. A farmer's good performance is known to a local collector, but it is not transferable to a new buyer. Without portable reputation, farmers cannot convert quality and reliability into bargaining power. DID and VC-based reputation addresses this issue by creating digital evidence of performance that is controlled by the farmer and verifiable by authorized buyers [18], [19].

The third gap is the lack of an integrated architecture that combines traceability, reputation, matching, aggregation, and smart contract-based settlement. Traceability alone records product movement, but it does not necessarily guarantee fair payment. A listing platform may connect parties, but it does not solve payment risk. A reputation system may improve confidence, but it must be linked to actual transaction execution. FreshRoute therefore fills the gap by proposing a holistic platform in which identity, reputation, order matching, aggregation, escrow, settlement, and traceability operate as mutually reinforcing components.

1.3 Research Problem

The research problem addressed by this dissertation is that the Sri Lankan fruit supply chain lacks a transparent, trusted, and technically accessible mechanism through which smallholder farmers and institutional buyers can transact directly based on verifiable identity, quality, and performance evidence. The current intermediary-led structure creates value loss through opaque pricing, delayed payments, high transaction dependency, weak quality incentives, and limited market power for farmers. Buyers also face difficulty in sourcing consistent quantities of quality produce from fragmented farmers without depending on existing intermediaries.

Although intermediaries perform useful functions, the absence of a digital alternative means that farmers remain vulnerable to unfair pricing and credit-bound selling arrangements. Existing digital platforms do not sufficiently replace the trust, aggregation, and payment assurance functions of intermediaries. Consequently, a system is required that can create verifiable trust between unknown parties, automate transaction enforcement, and support aggregation of produce from multiple smallholder farmers while remaining usable in a low digital literacy environment.

1.4 Research Questions

1. How can a blockchain-based direct marketplace be designed to be usable and adoptable by smallholder farmers in Sri Lanka, considering low digital literacy and limited access to technology?
2. To what extent can an on-chain reputation system built using Decentralized Identifiers and Verifiable Credentials replace the socio-economic trust functions of traditional intermediaries in the fruit supply chain?
3. What system architecture can effectively integrate traceability, verifiable reputation, intelligent matching, aggregation, and automated smart contracts into a holistic platform for improving market efficiency and equity?
4. What measurable economic and operational impact can the platform demonstrate on KPIs such as farmer share of final price, payment delay reduction, transaction transparency, and buyer trust during prototype evaluation?

1.5 Research Objectives

1.5.1 Main objective

The main objective of this research is to design, develop, and evaluate a blockchain-powered direct marketplace that leverages a verifiable on-chain reputation system to reduce intermediary dependency in the traditional Sri Lankan fruit supply chain. The system aims to improve transparency, encourage quality-based pricing, support timely settlement, and increase farmer bargaining power while enabling buyers to access reliable, traceable, and high-quality produce.

1.5.2 Specific objectives

1. To analyze the existing intermediary-led fruit market structure and identify the economic inefficiencies, trust relationships, pricing mechanisms, and operational challenges that disadvantage farmers and buyers.
2. To design a secure, private, and scalable system architecture using Hyperledger Fabric, including participant onboarding, end-to-end traceability, private transaction channels, and DID-based farmer identity.

3. To develop a functional prototype containing farmer supply listing, buyer order placement, an intelligent matching and aggregation engine, smart sales agreements, escrow-based settlement logic, and an on-chain reputation update mechanism.
4. To evaluate the feasibility, usability, performance, security, and potential economic impact of the prototype through controlled testing and pilot study procedures using KPIs such as farmer share of price, settlement time, successful match rate, transaction latency, and user trust.

1.6 Scope and Significance of the Study

The scope of the study is limited to the blockchain marketplace and digital trust component of the broader FreshRoute system. This includes participant registration, DID creation, on-chain reputation, supply and order management, matching and aggregation, smart contract transaction execution, traceability, and administrative governance. The study does not attempt to replace every physical function of the supply chain. Transportation, cold storage, crop forecasting, and quality assessment are considered related components that may be integrated with this module but are outside the main technical focus of this individual dissertation.

The significance of the study lies in its attempt to create a practical bridge between technological trust and agricultural market reform. For farmers, the system provides a means of converting quality and reliability into a verifiable asset. For buyers, it creates a lower-risk pathway to direct sourcing. For policymakers, the platform can generate anonymized and verified market intelligence. For researchers, the study contributes a context-specific model for applying permissioned blockchain, DIDs, VCs, and smart contracts to smallholder agriculture in a developing country environment.

1.7 Chapter Summary

This chapter established the background of the research by discussing the inefficiencies of Sri Lanka's fruit supply chain, the role of intermediaries, the issue of post-harvest loss, current modernization initiatives, and the potential of blockchain-based trust infrastructure. It identified a clear research gap at the intersection of smallholder agriculture, direct B2B marketplaces, and DID-based verifiable reputation. The research problem, research questions, objectives, scope, and significance were then defined to guide the remainder of the dissertation.

2 METHODOLOGY

2.1 Research Approach and Design

This research follows a mixed-methods design supported by the Design Science Research paradigm. The mixed-methods component is necessary because the problem is socio-technical: it involves farmer behavior, buyer risk perception, intermediary influence, payment practices, and technology adoption. Qualitative inquiry is used to understand stakeholder experiences and requirements, while design science is used to create and evaluate the IT artifact that addresses the identified problem.

Design Science Research is appropriate because the study aims to solve a real-world problem through the construction and evaluation of a purposeful technological artifact. In this dissertation, the artifact is the FreshRoute blockchain marketplace prototype. The research process therefore includes problem identification, objective definition, design and development, demonstration, evaluation, and communication. The artifact is evaluated not only by whether the software functions correctly, but also by whether its design addresses the trust and market access problems identified in the literature and stakeholder analysis.

The research is organized into five phases. Phase 1 focuses on problem identification and stakeholder requirement gathering. Phase 2 defines objectives and translates the problem into technical requirements. Phase 3 designs the system architecture, data model, and algorithms. Phase 4 implements the prototype using the selected technology stack. Phase 5 evaluates the artifact through functional testing, performance testing, usability testing, and pilot-study-based KPI analysis. [final details of actual evaluation sample size and dates after completion.]

2.2 Requirement Gathering and Analysis

Requirement gathering was designed to capture the needs of the three most important stakeholder groups: smallholder farmers, institutional buyers, and intermediaries. Farmers are the primary beneficiaries because the platform is expected to increase their access to transparent market opportunities. Buyers are the primary paying customers because they receive traceability, verified supplier access, reduced counterparty risk, and procurement data. Intermediaries are included in the analysis because they perform functions that must either be replaced, redesigned, or integrated into a more transparent ecosystem.

Semi-structured interviews were selected as the primary method because they allow consistent coverage of core research themes while leaving room for participants to describe context-specific problems. Farmer interviews focus on pricing, payment cycles, credit dependency, transport barriers, smartphone access, and trust in digital systems. Buyer interviews focus on procurement quantities, quality requirements, traceability needs, supplier risk, and direct sourcing barriers. Intermediary interviews focus on cost structures, aggregation processes, credit provision, and perceived operational value.

The interview data is intended to be analyzed through thematic analysis. The process includes familiarization with transcripts, initial coding, theme development, theme review, and mapping themes to system requirements. For example, if farmers repeatedly identify delayed payment as a primary pain point, this theme is mapped to the Smart Sales Agreement escrow requirement. If buyers identify quality inconsistency as a major risk, this theme is mapped to quality-linked Verifiable Credentials and reputation scoring. This approach ensures that technical features are grounded in actual stakeholder needs rather than being included only for technological novelty.

2.2.1 Functional requirements

Table 5: Functional Requirements

ID	Requirement	Description
FR1	Secure user onboarding and identity management	Verified farmers, buyers, and administrators must be registered, approved, and assigned unique DIDs.
FR2	On-chain reputation management	The platform must update farmer reputation after verified transactions using delivery, quality, and feedback evidence.

ID	Requirement	Description
FR3	Farmer supply listing	Farmers must be able to list fruit type, quantity, grade, expected harvest date, price, and location.
FR4	Buyer order placement	Buyers must create orders with product, quantity, grade, maximum price, delivery date, and destination.
FR5	Intelligent matching and aggregation	The engine must rank and combine farmers to satisfy buyer demand based on reputation, price, quantity, and proximity.
FR6	Smart contract automation	A Smart Sales Agreement must manage escrow, delivery confirmation, and settlement state transitions.
FR7	Traceability and transaction visibility	Authorized transaction participants must view order status and audit history from the ledger.
FR8	Administrative governance	Administrators must verify users, monitor network activity, resolve disputes, and manage platform governance rules.

2.2.2 Non-functional requirements

Table 6: Non-Functional Requirements

Category	Requirement
Security	End-to-end communication security, certificate-based identity, role-based access control, private key protection, and confidential transaction handling.
Scalability	The architecture must allow additional peer nodes, organizations, buyers, and farmers without redesigning the entire network.
Usability	Farmer-facing screens must use simple workflows, minimal text input, clear language, and visual cues suitable for varying digital literacy levels.

Category	Requirement
Reliability	The platform should be designed for high availability through redundant peers and cloud-based deployment.
Performance	Read queries should target less than 2 seconds, while write transactions should target commitment within 5 to 10 seconds under prototype load conditions.
Immutability	Completed transactions and reputation events must be cryptographically protected against unauthorized alteration.
Transparency	The logic of reputation scoring and smart contract settlement should be auditable by authorized participants.
Privacy	Sensitive commercial data such as prices and buyer orders must be protected through Fabric channels, private data collections, and access policies.

2.3 Feasibility Study

2.3.1 Technical feasibility

The proposed system is technically feasible because the required components are supported by mature open-source technologies. Hyperledger Fabric provides a permissioned blockchain architecture with membership services, chaincode execution, channels, and private data collections. JavaScript is a suitable language for chaincode because it is compiled, statically typed, and widely used in Fabric examples. Node.js and Express.js provide a practical middleware layer between the user interfaces and the blockchain network through Fabric SDK integration. CouchDB supports rich JSON queries over world-state data, allowing efficient filtering of farmer supply listings and buyer orders. IPFS provides a practical off-chain storage method for images, quality certificates, and supporting documents, while storing only file hashes on-chain.

The main technical challenge is integration. A blockchain prototype is not only a database; it requires coordination among peer nodes, orderers, certificate authorities, endorsement policies, chaincode lifecycle management, API authentication, and client application workflows. This challenge is manageable within the dissertation scope by implementing a limited but complete prototype network with representative organizations such as Farmers, Buyers, Quality Assessors, and Admin/Governance.

2.3.2 Operational feasibility

Operational feasibility depends on adoption by farmers and buyers. Farmers may have limited digital literacy and inconsistent access to smartphones or stable connectivity. Therefore, the farmer application must be mobile-first, simple, and tolerant of low text input. Buyers, in contrast, require detailed dashboards, order management, traceability, and analytics. The system is operationally feasible if the two user experiences are designed separately while using the same underlying trusted data layer.

The platform also requires human governance. Administrators or authorized field officers must verify farmers, resolve disputes, and validate certain credentials when automated confirmation is not sufficient. This means the platform should be understood as a socio-technical ecosystem rather than a fully autonomous replacement for all human judgment.

2.3.3 Economic feasibility

The projected development and pilot budget in the proposal is LKR 50,000, consisting of cloud services, software tools, internet cost, and field research expenses. The use of open-source technologies reduces licensing cost. Commercial sustainability is expected through transaction commissions charged to buyers, premium subscription tiers for high-volume buyers, API access, enhanced analytics, priority matching, and anonymized market intelligence reports. Since farmer adoption is essential to supply network growth, farmer access should remain free or heavily subsidized during early stages.

2.3.4 Legal, ethical, and privacy feasibility

The system handles identity, commercial, and transaction data. Therefore, privacy and consent must be central to implementation. Only verified participants should join the network, and access to sensitive information must be limited according to role and transaction relationship. Farmers must understand how their reputation score is calculated, how their data is used, and what rights they have regarding correction of errors. While blockchain immutability strengthens auditability, it also requires careful data design: personal documents and large evidence files should not be stored directly on-chain. Instead, the prototype stores hashes and references, while the files themselves are managed through IPFS or secure off-chain storage.

2.4 System Design

2.4.1 Overall system architecture

The FreshRoute blockchain marketplace is designed as a multi-layered system. The presentation layer consists of a farmer mobile application, buyer web portal, and administrator dashboard. The application layer consists of Node.js and Express.js APIs that validate requests, interact with the Fabric SDK, manage authentication sessions, and coordinate off-chain file handling. The blockchain layer consists of Hyperledger Fabric peers, ordering service, certificate authorities, channels, chaincode, CouchDB state databases, and private data collections. The storage layer includes the blockchain ledger for immutable transaction evidence and IPFS for large files such as quality images and certificates.

The architecture separates user experience from trust execution. Users interact with simple screens and dashboards, while business rules are enforced by smart contracts. This approach reduces the need for users to understand blockchain details. A farmer lists harvest details through the mobile app; the backend translates this request into a Fabric transaction; chaincode validates the user role and writes a supply listing to the ledger state. A buyer places an order; the matching engine queries CouchDB, ranks eligible supply, and creates a fulfillment plan. When both parties accept, the Smart Sales Agreement records escrow and delivery state transitions.

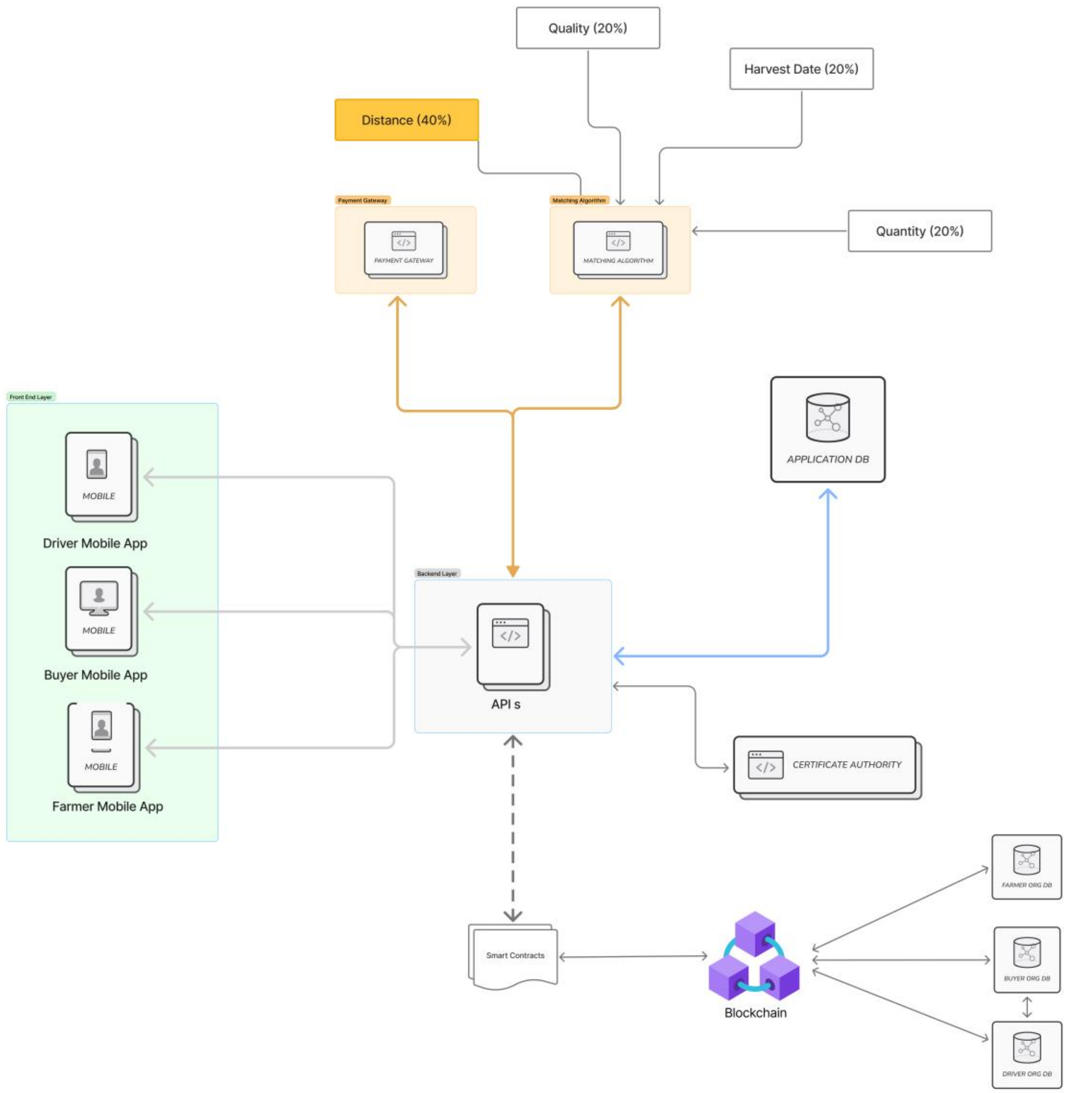

Figure 2: Overall Architecture Diagram

2.4.2 Hyperledger Fabric network design

The Fabric network is designed around known organizations. A prototype deployment may include a Farmer Organization, Buyer Organization, Quality Assessor Organization, and Platform Admin Organization. Each organization operates or is represented by peer nodes and has identities issued by a Certificate Authority. The Membership Service Provider defines trusted certificate roots and identity rules.

Endorsement policies specify which organizations must endorse particular transactions. For example, a delivery confirmation may require endorsement from the farmer, buyer, and quality assessor before settlement is triggered.

Channels and private data collections are used to preserve confidentiality. General reputation metadata can be visible to authorized buyers, while specific price agreements should be visible only to the transacting parties and required governance actors. This design is important because buyers may not be willing to participate if commercially sensitive procurement data is exposed to competitors. Hyperledger Fabric is therefore more suitable than public blockchains for this B2B use case.

Figure 3: Hyperledger Fabric Network Topology

2.4.3 Core data model

The blockchain data model is designed around assets and transaction events. Assets represent current business objects such as farmer profiles, supply listings, buyer orders, and trade agreements. Events represent changes in state such as listing creation, order placement, match generation, escrow activation, delivery confirmation, settlement, and reputation update. World-state records in CouchDB support efficient queries, while the blockchain ledger preserves the immutable history of all committed changes.

Table 7: Core Data Model of the Blockchain Marketplace

Asset / Record	Key fields	Purpose
UserIdentity	userID, DID, role, organization, certificateRef, status	Represents verified participants and links platform accounts to blockchain identities.
FarmerProfile	farmerID, DID, crops, location, cooperativeID, reputationScore	Stores farmer attributes required for matching and reputation evaluation.
SupplyListing	listingID, farmerID, productID, quantity, grade, pricePerKg, availableDate, location	Represents available or predicted supply listed by farmers.

Asset / Record	Key fields	Purpose
BuyerOrder	orderID, buyerID, productID, requiredQuantity, grade, maxPrice, deliveryDate, destination	Represents buyer demand submitted to the marketplace.
TradeAgreement	agreementID, orderID, farmerIDs, quantities, price, escrowStatus, deliveryStatus, settlementStatus	Represents the smart sales agreement and its lifecycle.
ReputationRecord	recordID, farmerID, deliveryScore, qualityScore, feedbackScore, timestamp	Stores the evidence used to update farmer reputation.
CredentialReference	credentialID, DID, claimHash, issuer, IPFSHash, issueDate	Links verifiable credential evidence to DID and off-chain files.
DisputeRecord	disputeID, agreementID, raisedBy, reason, status, resolution	Supports governance and dispute resolution.

2.4.4 Security and privacy design

Security is addressed at multiple layers. At the network level, Hyperledger Fabric uses certificate-based identities issued by Certificate Authorities. At the communication level, TLS should be enabled between clients, APIs, and blockchain network components to protect data in transit. At the application level, role-based access control ensures that farmers, buyers, administrators, and quality assessors can only perform actions permitted by their roles. At the data layer, private data collections and channels restrict access to sensitive price and transaction details.

Private keys must be protected because they represent participant authority in the blockchain network. In the prototype, keys may be stored in a secure development wallet managed by the backend. In a production environment, a mobile wallet, hardware-backed key storage, or managed custody model would be required to balance user control and usability. Since many farmers may not be familiar with private key management, the system must avoid exposing cryptographic complexity directly to end users.

2.5 Implementation

2.5.1 Development environment and tools

The implementation uses Hyperledger Fabric as the blockchain framework, Go for smart contracts, Node.js with Express.js for the backend API, CouchDB as the Fabric state database, IPFS for off-chain storage, React Native for the farmer mobile application, and React.js for the buyer and administrator web portals. The development environment includes Docker containers for Fabric peers, orderers, certificate authorities, and CouchDB instances. Git is used for version control, and API endpoints are tested using standard HTTP clients. [repository link, branch structure, and deployment screenshots.]

Table 8: Technology Stack Justification

Component	Technology	Justification
Blockchain platform	Hyperledger Fabric	Permissioned, enterprise-grade, suitable for privacy-preserving B2B supply chain networks.
Smart contracts	JavaScript	Compiled, statically typed, performant, and appropriate for secure chaincode logic.
Backend API	Node.js with Express.js	Supports asynchronous API workflows and Fabric SDK integration.
Frontend	React Native and React.js	Supports mobile farmer access and responsive web dashboards for buyers and administrators.
State database	CouchDB	Stores JSON world state and enables rich queries for matching and filtering.

Component	Technology	Justification
Off-chain storage	IPFS	Stores large files while blockchain stores hashes for tamper evidence.
Security	TLS, MSP, RBAC	Protects transport, identity, and permission boundaries.

2.5.2 Smart contract implementation

The chaincode layer implements the core marketplace rules. The UserIdentity contract registers verified participants and links each profile with a DID. The SupplyListing contract allows farmers to create, update, and deactivate supply records. The BuyerOrder contract records buyer demand and validates order constraints. The Matching contract stores the generated fulfillment plan after the off-chain matching engine calculates the recommended farmer group. The SmartSalesAgreement contract manages escrow status, delivery confirmation, quality verification, dispute status, and settlement completion.

The smart contract functions are designed to be deterministic. Blockchain chaincode must produce the same output on all endorsing peers for the same input. Therefore, non-deterministic operations such as random selection, live external API calls, and time-dependent calculations are avoided inside chaincode. The matching score calculation can be performed by the backend engine using the latest world-state data, and the resulting plan is then submitted to chaincode for validation and recording. This separation keeps chaincode deterministic while still allowing the matching engine to use advanced algorithms.

Example chaincode functions include RegisterFarmer(), CreateDIDRecord(), CreateSupplyListing(), CreateBuyerOrder(), SubmitMatchProposal(), CreateSmartSalesAgreement(), ConfirmEscrow(), ConfirmDelivery(), ConfirmQualityGrade(), ReleaseSettlement(), RaiseDispute(), ResolveDispute(), and UpdateReputation(). [code screenshots for major chaincode functions.]

2.5.3 Decentralized identity and verifiable reputation implementation

The DID implementation provides a stable digital identity for each verified farmer and buyer. During onboarding, the administrator verifies the participant and triggers DID creation. The DID document contains public key material and service endpoints relevant to the platform. The private key remains under controlled storage and is used to sign relevant transactions or credential requests according to the prototype design.

Verifiable reputation is implemented as a weighted score derived from transaction outcomes. A simple prototype formula can combine delivery performance, quality grade consistency, buyer feedback, dispute history, and settlement completion. For example, the score may be calculated as: Reputation = (0.40 x deliveryScore) + (0.35 x qualityScore) + (0.15 x buyerFeedbackScore) + (0.10 x disputePenaltyAdjustedScore). The exact weights should be validated with domain experts and may differ for export-focused buyers, local supermarkets, and processors.

Each completed transaction can produce a credential reference. The credential itself may include the transaction identifier, product, quantity, grade, delivery date, issuer, and cryptographic proof. Storing a hash of the credential on-chain allows future verification without storing all personal or commercial details directly on the ledger. This design maintains a balance between immutability and privacy.

2.5.4 Intelligent matching and aggregation algorithm

The Intelligent Matching and Aggregation Engine is the core algorithmic component of the marketplace. Its purpose is to transform fragmented smallholder supply into a fulfillment plan that can satisfy institutional buyer demand. The algorithm begins when a buyer submits an order specifying product type, required quantity, quality grade, latest delivery date, maximum acceptable price, and delivery location. The backend queries the CouchDB world state for all active supply listings that satisfy the hard constraints.

After filtering, each eligible farmer is assigned a match score. The proposed scoring formula is Match_Score = (wrep x Reputation_Score) + (wprice x Price_Score) + (wloc x Location_Score). The reputation score reflects farmer reliability and quality history. The price score rewards listings that fall below the buyer's maximum price. The location score encourages logistical efficiency by prioritizing farmers closer to the delivery point or to other high-scoring farmers. The weights are configurable. For export orders, reputation may be prioritized; for processing orders, price and quantity may receive higher weight.

Once scores are calculated, farmers are sorted in descending order. The aggregation loop then selects farmers until the required quantity is satisfied or no eligible supply remains. A greedy approach is used in the prototype because it is explainable, simple, and efficient. Future versions may use integer programming or metaheuristic optimization to minimize total cost while satisfying quality and route constraints.

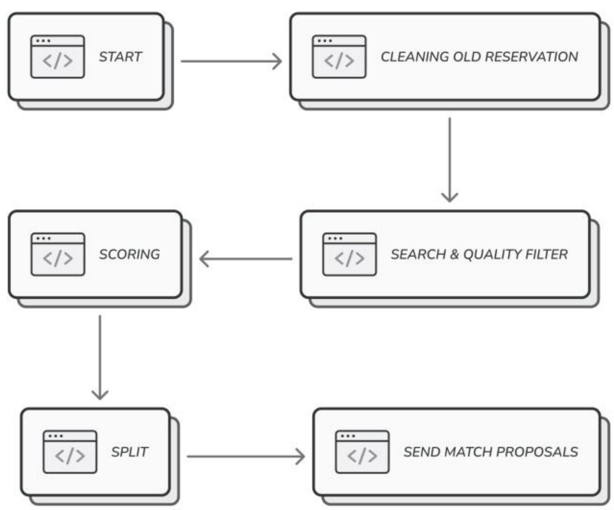

Figure 4: Matching Algorithm Flow

The prototype pseudocode is as follows: receive BuyerOrder; query eligible SupplyListings; remove listings that do not meet product, grade, date, and price constraints; normalize reputation, price, and location scores; compute match score; sort farmers; select listings until required quantity is reached; generate fulfillment group; present proposal to buyer; on confirmation, create a Master Smart Sales Agreement and linked sub-contracts for each participating farmer. [Actual pseudocode/code listing in Appendix after implementation.]

2.5.5 Backend API and user interface implementation

The backend API acts as the bridge between user interfaces and the blockchain network. It authenticates users, validates form input, invokes chaincode transactions, submits ledger queries, manages file uploads to IPFS, and returns user-friendly responses. The API endpoints include farmer registration, admin approval, supply listing creation, buyer order placement, matching request, agreement confirmation, delivery confirmation, quality confirmation, reputation query, and dispute management.

The farmer mobile interface is designed around simplified workflows: register, view profile and reputation, create supply listing, view matched orders, confirm delivery status, and review payment status. The buyer web interface supports order creation, match review, supplier reputation inspection, traceability viewing, and settlement monitoring. The administrator dashboard supports participant approval, transaction monitoring, dispute handling, and system health review.

Figure 5: Farmer Onboarding and DID Creation Interface

Figure 6: Buyer Order Placement and Matching Interface

2.5.6 Integration with the broader FreshRoute system

This dissertation focuses on the blockchain marketplace component, but the wider FreshRoute system includes related components such as AI demand forecasting, spoilage-reduction logistics, and quality assurance. Integration points are therefore defined at the API and data levels. Demand forecasts can inform farmer supply planning and buyer procurement recommendations. Quality assurance outputs can act as credential issuers for grade verification. Logistics optimization can use confirmed smart sales agreements to plan transport routes. The blockchain marketplace functions as the trusted transaction and reputation layer that records verified outcomes from these components.

2.6 Testing and Validation

Testing is planned across multiple levels to validate both technical correctness and user acceptance. Unit testing verifies individual smart contract functions and backend services. Integration testing verifies interactions among the frontend, backend API, Fabric network, CouchDB, and IPFS. System testing validates complete workflows such as onboarding, listing, matching, agreement creation, delivery confirmation, and reputation update. Performance testing measures query response time and transaction commitment time under simulated load. Security testing verifies access control, unauthorized role actions, and confidentiality of private transaction data. Usability testing assesses whether farmers and buyers can complete critical workflows with minimal assistance.

Table 9: Technology Stack Justification

Test ID	Test area	Objective	Expected outcome
TC01	User onboarding and DID creation	Verify that an approved user receives a unique DID linked to the ledger profile.	DID is created and queryable by authorized parties.
TC02	Smart contract execution	Validate escrow activation, delivery confirmation, and automated settlement.	Agreement state transitions occur correctly without manual ledger changes.
TC03	Reputation score update	Ensure completed transactions, update farmer reputation and create credential reference.	Reputation changes according to scoring rules and VC hash is recorded.

Test ID	Test area	Objective	Expected outcome
TC04	Matching and aggregation	Confirm that multiple farmers can be selected to fulfill one buyer order.	Fulfillment plan satisfies quantity and prioritizes high-scoring farmers.
TC05	Data privacy	Verify that confidential transaction data is not visible to unauthorized roles.	Unauthorized query attempts are rejected or return restricted data.
TC06	Performance	Measure read and write transaction response times under simulated concurrent users.	Results remain within target thresholds or are documented for optimization.
TC07	Usability	Assess whether farmers can complete registration and listing workflows.	Task completion rate and satisfaction meet target benchmarks.

2.7 Commercialization Aspects of the Product

The commercialization vision of FreshRoute is to become the digital trust layer for Sri Lanka's fruit supply chain. The platform's market opportunity arises from the persistent lack of transparency, payment assurance, traceability, and direct access between farmers and buyers. By establishing a trusted marketplace, the system creates value for multiple stakeholder groups. Buyers gain access to verified farmers and traceability data. Farmers gain direct market visibility, timely payment, and reputation-based bargaining power. Government and development organizations gain access to aggregated market intelligence that can support policy interventions.

The target market follows a multi-sided B2B model. Primary paying customers are exporters, processors, supermarket chains, and high-volume procurement organizations. Farmers are treated as the supply network and primary beneficiaries; their access should remain free during early adoption to reduce barriers. Ecosystem partners include banks, microfinance institutions, insurers, logistics providers, certification bodies, and agricultural extension services. The platform can eventually provide verified reputation data for credit scoring, insurance pricing, and quality assurance.

Revenue streams include a transaction commission charged to buyers for completed trades, premium SaaS subscription tiers for high-volume buyers, API access for procurement integration, enhanced farmer reputation analytics, priority matching, and anonymized market intelligence reports. The go-to-market strategy is staged. Year 1 should validate the trust engine through a focused pilot involving a limited crop category and selected buyer group. Years 2 and 3 should scale farmer and buyer onboarding through farmer producer organizations and agricultural extension networks. Year 4 onward can open APIs for integration with finance, logistics, and policy analytics partners.

The major competitive advantage is the verifiable trust network. Traditional marketplaces can be copied, but a growing ledger of verified farmer reputation and transaction history becomes increasingly valuable as adoption grows. Additional advantages include tamper-evident data integrity, intelligent aggregation for large orders, alignment with national modernization goals, and the ability to integrate other FreshRoute components.

Table 10: Budget Table

Item	Estimated Cost (LKR)	Justification
Cloud services	25,000.00	Deployment of blockchain networks, APIs, and testing environments.
Software and tools	5,000.00	Development utilities and supporting tools are not available through free tiers.
Internet cost	5,000.00	Connectivity for development, deployment, and stakeholder communication.

Item	Estimated Cost (LKR)	Justification
Field research and miscellaneous	15,000.00	Travel, interviews, printing, and pilot study expenses.
Total	50,000.00	Projected development and pilot budget.

2.8 Chapter Summary

This chapter described the methodology used to design and evaluate the FreshRoute blockchain marketplace. It explained the mixed-methods and design science approach, stakeholder requirement gathering, feasibility analysis, system architecture, Hyperledger Fabric network design, data model, security architecture, implementation plan, testing strategy, and commercialization approach. The methodology establishes a clear pathway from the identified research problem to a technically implementable and commercially relevant solution.

3 RESULTS AND DISCUSSION

3.1 Results

This section presents the expected structure for reporting the implementation and evaluation results of the FreshRoute blockchain marketplace. Since this document is prepared as a final thesis draft based on the project proposal, numerical values and screenshots that depend on actual implementation and pilot testing are represented using placeholders. These placeholders must be replaced with final results after system testing, user evaluation, and pilot study completion.

3.1.1 Prototype implementation results

The prototype implementation is expected to demonstrate the end-to-end workflow of the direct marketplace. The first workflow is secure onboarding, where a farmer submits registration details and an administrator approves the account. After approval, a DID is generated and linked with the farmer profile. The second workflow is supply listing, where the farmer records fruit type, quantity, grade, expected date, price, and location. The third workflow is buyer order placement, where an institutional buyer specifies demand details. The fourth workflow is matching and aggregation, where the algorithm generates a fulfillment group from multiple farmers. The fifth workflow is smart sales agreement execution, where the buyer confirms the fulfillment group and the system records escrow and settlement states. The sixth workflow is reputation update, where successful delivery and quality verification update the farmer's score and create a credential reference.

[Screenshots of the completed prototype workflow: registration page, DID creation confirmation, supply listing page, buyer order page, match result page, smart contract transaction log, and reputation dashboard.]

Table 11: Prototype Result Matrix

Function	Expected result	Actual result	Status
Farmer onboarding	Approved farmers receive DID and active profile.	[Actual result]	[Pass/Fail]

Function	Expected result	Actual result	Status
Supply listing	Farmer can create a valid harvest listing.	[Actual result]	[Pass/Fail]
Buyer order	Buyer can submit order constraints.	[Actual result]	[Pass/Fail]
Matching	System returns ranked and aggregated farmer group.	[Actual result]	[Pass/Fail]
Smart sales agreement	Escrow and delivery states are recorded correctly.	[Actual result]	[Pass/Fail]
Reputation update	Reputation score and credential hash are updated.	[Actual result]	[Pass/Fail]
Traceability	Authorized users can view transaction history.	[Actual result]	[Pass/Fail]

3.1.2 Performance testing results

Performance testing focuses on read query time, write transaction commitment time, matching execution time, and API response time under increasing concurrency. Read operations include fetching farmer reputation, listing available supply, and viewing transaction status. Write operations include creating listings, submitting orders, confirming delivery, and updating reputation. The target defined in the proposal is that read queries should complete in under 2 seconds and write transactions should commit within 5 to 10 seconds under prototype conditions.

Figure 7 : Performance Testing Result Graph

Table 12: KPI Evaluation Plan and Placeholder Results

KPI	Measurement method	Target / Expected value	Actual value
Read query latency	Average time for reputation or listing query	< 2 seconds	
Write transaction latency	Average time to commit listing/order/settlement	5-10 seconds	
Successful match rate	Orders matched with available supply / total orders	[target]	
Farmer share of final price	Farmer received price / comparable final buyer price	Improvement over baseline	
Payment delay reduction	Baseline delay minus platform settlement time	Significant reduction	
User task completion	Completed usability tasks / total assigned tasks	>= [target]%	
User trust score	Survey score before and after prototype demonstration	Improvement over baseline	

3.1.3 Usability and stakeholder feedback results

Usability evaluation should be reported separately for farmers and buyers because their technical expectations differ. Farmer evaluation should focus on registration, supply listing, viewing matched orders, and understanding payment status. Buyer evaluation should focus on order placement, interpreting reputation scores, reviewing fulfillment plans, and viewing traceability. The evaluation may use task completion rate, time-on-task, error count, and post-test satisfaction score. Qualitative feedback should be coded into themes such as trust, simplicity, perceived fairness, payment confidence, and willingness to adopt.

3.1.4 Economic impact evaluation results

The economic impact of the platform should be evaluated cautiously because a prototype pilot cannot fully represent national market dynamics. Nevertheless, a controlled comparison can estimate potential improvement in farmer share of final price, reduction in payment delays, and reduction in transaction uncertainty. Baseline values can be obtained from literature, stakeholder interviews, or current market transactions. Platform values can be obtained from simulated or pilot trades. The final report should clearly state whether the values are simulated, pilot-based, or calculated from stakeholder-provided estimates.

3.2 Research Findings

The first finding is that trust is the central barrier to direct farmer-buyer trade. Farmers require assurance that buyers will pay fairly and on time, while buyers require assurance that farmers can deliver the required quantity and quality. A simple marketplace cannot resolve this issue unless it includes mechanisms for verifiable identity, transaction accountability, and reputation.

The second finding is that Hyperledger Fabric is technically aligned with the privacy needs of B2B agricultural trade. Public blockchains provide transparency but expose data in ways that may be unsuitable for commercial procurement. Fabric's permissioned participation, channels, private data collections, and endorsement policies provide a stronger fit for known participants who require accountability and confidentiality.

The third finding is that DID and VC-based reputation can conceptually replace part of the trust function performed by intermediaries. Intermediaries traditionally validate farmer reliability through repeated local relationships. FreshRoute records verified transaction outcomes and converts them into portable digital evidence. This does not eliminate the need for all human relationships, but it reduces dependence on informal gatekeepers.

The fourth finding is that aggregation is essential for institutional buyer adoption. Large buyers rarely purchase the small quantities produced by individual smallholders. The matching and aggregation engine therefore plays a crucial role by converting fragmented supply into a buyer-ready fulfillment plan.

The fifth finding is that usability is as important as technical correctness. A technically secure blockchain platform will fail if farmers cannot understand the interface or if key management creates excessive cognitive burden. The system must hide cryptographic complexity behind simple workflows, clear status indicators, and assisted onboarding.

3.3 Discussion

3.3.1 Discussion of research questions

RQ1 asked how a blockchain-based direct marketplace can be designed for smallholder farmer usability. The proposed design addresses this through a mobile-first farmer interface, minimal text input, administrator-assisted verification, simple listing forms, and clear transaction status messages. The blockchain layer is hidden from the user interface so that farmers interact with familiar marketplace concepts rather than cryptographic operations.

RQ2 asked whether DID and VC-based on-chain reputation can replace intermediary trust functions. The system can replace the evidence and verification component of trust by creating tamper-evident records of identity, quality, delivery, and settlement. However, it may not fully replace all intermediary functions such as physical aggregation, credit provision, and local conflict mediation. Therefore, the platform should be viewed as a disintermediation and re-intermediation tool: it reduces exploitative dependency while allowing transparent service providers to participate under auditable rules.

RQ3 asked what architecture can integrate traceability, reputation, and smart contracts. The proposed architecture uses Hyperledger Fabric as the ledger and governance layer, Go chaincode as the business rule layer, CouchDB for queryable world state, IPFS for off-chain evidence, Node.js APIs for integration, and React/React Native interfaces for user access. This architecture enables modular development and provides the privacy controls required for B2B trading.

RQ4 asked what measurable impact can be observed through evaluation. The final answer must be completed after pilot testing. The draft evaluation framework measures farmer share of final price, payment delay reduction, match success rate, read/write latency, task completion, and user trust.

3.3.2 Implications

The research has practical implications for farmers, buyers, policymakers, and technology developers. For farmers, a verifiable reputation system can transform repeated good performance into a market asset. For buyers, the system reduces uncertainty in direct sourcing and supports traceability requirements. For policymakers, aggregated platform data can reveal price trends, supply gaps, and bottlenecks. For software engineers, the research demonstrates how blockchain should be applied selectively as a trust and audit layer rather than as a replacement for all database functions.

3.3.3 Limitations and challenges

Several limitations must be acknowledged. First, the prototype may be evaluated using a limited pilot sample, which restricts generalizability. Second, farmer smartphone access and digital literacy may affect adoption. Third, blockchain immutability requires careful handling of incorrect data; dispute and correction workflows must be designed without deleting history. Fourth, physical verification of quality remains a challenge unless integrated with trusted assessors, IoT devices, or quality assurance partners. Fifth, the platform cannot eliminate all intermediary roles because transport, storage, grading, and credit remain real operational needs. The goal is therefore not to remove every actor but to eliminate opaque and exploitative dependency.

3.3.4 Future work

Future work should extend the prototype into a larger field pilot involving multiple districts, fruit types, and buyer categories. The matching algorithm can be enhanced using route optimization, dynamic pricing, and predictive demand signals. Reputation scoring should be validated with domain experts to avoid unfair bias against new farmers. Integration with digital payment providers, microfinance institutions, crop insurance, and quality certification services would strengthen commercialization potential. Finally, offline-first mobile capabilities and Sinhala/Tamil language support should be added to improve adoption among rural users.

3.5 Chapter Summary

This chapter presented the results reporting structure, expected prototype outputs, KPI evaluation plan, major research findings, discussion of research questions, implications, limitations, future work, and contribution summary. The placeholders in this chapter should be replaced with actual system screenshots, test logs, pilot results, and stakeholder feedback after implementation and evaluation are complete.

4 CONCLUSION

This dissertation presented FreshRoute, a blockchain-powered direct marketplace designed to reduce waste, increase transparency, and reduce exploitative intermediary dependency in Sri Lanka's fruit supply chain. The research was motivated by persistent post-harvest losses, opaque pricing, delayed payments, limited farmer bargaining power, and buyer difficulty in sourcing reliable produce directly from fragmented smallholder farmers. The study identified that the core challenge is not merely a lack of digital listing platforms but a deeper lack of verifiable trust between unknown market participants.

The proposed solution addresses this challenge by combining Hyperledger Fabric, smart contracts, Decentralized Identifiers, Verifiable Credentials, intelligent matching and aggregation, and off-chain evidence storage. Hyperledger Fabric provides a permissioned and privacy-preserving network suitable for B2B supply chain transactions. Smart Sales Agreements automate transaction states such as escrow, delivery confirmation, quality verification, and settlement. DID-based identity and VC-based reputation allow farmers to build portable digital trust from verified performance. The matching engine aggregates supply from multiple smallholders so that institutional buyers can source required quantities without depending exclusively on traditional intermediaries.

The research contributes to software engineering and agricultural supply chain modernization by presenting a socio-technical architecture that is grounded in the Sri Lankan smallholder context. It demonstrates how blockchain can be used as a trust infrastructure rather than as a generic technology trend. The platform is designed to preserve privacy, support scalability, and remain usable for participants with varying levels of digital literacy. Its commercial potential lies in transaction commissions, buyer subscriptions, analytics services, and ecosystem partnerships with finance, logistics, and quality assurance providers.

The final effectiveness of the system must be validated through implementation evidence and pilot evaluation. The thesis draft therefore includes placeholders for screenshots, performance graphs, actual test outcomes, stakeholder feedback, and economic impact metrics. Once these results are inserted, the dissertation will provide a complete account of whether the FreshRoute blockchain marketplace can improve farmer share of value, reduce payment delays, increase buyer trust, and support a more transparent fruit supply chain in Sri Lanka.

In conclusion, FreshRoute provides a feasible and context-sensitive pathway toward a more equitable agricultural marketplace. It does not claim that technology alone can solve all structural problems in the supply chain. However, by converting identity, quality, delivery, and settlement into verifiable digital evidence, the system can reduce information asymmetry and create a foundation for fairer, more resilient, and more efficient agricultural trade.

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APPENDICES

Appendix A: Sample Farmer Interview Questionnaire

1. Name of farmer
2. Location (district and village)
3. Age
4. Total farm size in acres
5. Primary fruit crops cultivated
6. Years of farming experience
7. Primary buyer or sales channel
8. How the price is determined and who sets it
9. Payment method and frequency of delays
10. Biggest challenges when selling produce
11. Relationship with regular collectors or traders
12. Whether intermediaries provide loans, seeds, or fertilizer
13. Perception of fairness in current prices
14. Experience with direct selling
15. Smartphone ownership and mobile application comfort
16. Interest in a direct digital marketplace
17. Concerns about trust, complexity, transport, or payment
18. Most important features expected in a digital platform

Appendix B: Sample Buyer Interview Questionnaire

1. Company name and respondent role
2. Type of business
3. Primary fruits procured from Sri Lanka
4. Annual volume requirements

5. Current sourcing method

6. Challenges in procuring required quality and quantity

7. Quality and safety verification process

8. Main supply chain risks

9. Experience or interest in direct sourcing from smallholders

10. Expected benefits of verified direct sourcing

11. Importance of farm-to-fork traceability

12. Value of verifiable farmer performance records

13. Essential features required in a digital procurement platform

Appendix C: Detailed Test Case Template

Field	**Description**
Test case ID	[ID]
Module	[module]
Objective	[objective]
Preconditions	[preconditions]
Test steps	[numbered steps]
Expected result	[expected output]
Actual result	[actual output]
Status	Pass / Fail
Remarks	[notes]

Appendix D: Placeholder for System Screenshots

[Screenshots of the implemented application: farmer mobile app, buyer web portal, admin dashboard, blockchain transaction logs, IPFS file hash record, and performance testing dashboard.]

Appendix E: Ethics, Consent, and Pilot Study Documentation

[Participant information sheet, consent form, data privacy notice, ethics approval, dataset approval, pilot study attendance, and stakeholder validation letters.]

Appendix F: Draft User Guide for Prototype Demonstration

This appendix provides a short operational guide that can be used during the prototype demonstration and user acceptance testing sessions. The guide should be replaced or expanded with final screenshots after the implementation has been completed. It is included in the draft report to ensure that the final dissertation demonstrates not only system design but also practical usability.

F.1 Farmer workflow

1. Open the FreshRoute farmer mobile application and select the registration option.
2. Enter basic profile details such as name, district, village, crop type, and contact number. The final implementation should support Sinhala and Tamil labels where possible.
3. Submit the registration request and wait for administrator verification. The farmer should not be asked to manage blockchain keys manually during this step.
4. After approval, view the profile page and confirm that the system displays an active account status and a generated DID reference.
5. Select Create Supply Listing and enter fruit type, estimated quantity, quality grade, expected harvest date, asking price per kilogram, and farm location.
6. Review the listing summary and submit it. The application should display a confirmation message after the transaction has been committed to the ledger.
7. When a buyer order is matched, open the matched order notification and review quantity, price, expected delivery date, and payment status.
8. After delivery, verify that the transaction status changes according to the smart sales agreement workflow and that reputation score updates are reflected in the profile dashboard.

F.2 Buyer workflow

1. Open the FreshRoute buyer web portal and log in with verified buyer credentials.
2. Create a purchase order by selecting product type, required quantity, grade, delivery date, delivery destination, and maximum price.

3. Run the matching process and review the proposed fulfillment group generated by the intelligent matching and aggregation engine.
4. Inspect farmer reputation scores, available quantities, locations, and traceability references before confirming the order.
5. Confirm the fulfillment proposal. The system should create a master smart sales agreement and linked farmer sub-agreements.
6. Monitor escrow status, delivery status, quality verification, and settlement completion through the order dashboard.
7. Download or view traceability and reputation evidence for internal procurement records. In the final system, this output may be exported as a report or integrated through an API.

Appendix G: Risk Register and Mitigation Plan

The following risk register summarizes major technical, operational, commercial, and ethical risks associated with the FreshRoute blockchain marketplace. The final report should update this table with actual issues encountered during development and pilot testing.

Risk ID	Risk category	Risk description	Potential impact	Mitigation strategy
R1	Technical	Difficulty configuring Fabric channels, endorsement policies, or chaincode lifecycle.	Delayed implementation and incomplete prototype workflows.	Use a minimal viable network first, document configuration steps, and expand policies gradually.
R2	Usability	Farmers may find digital onboarding or listing forms difficult.	Low adoption and inaccurate data entry.	Use assisted onboarding, simple interfaces, visual cues, local language support, and field officer support.
R3	Security	Private keys or credentials may be mishandled by users.	Unauthorized transactions or loss of account access.	Use managed wallets during pilot and design a secure recovery process for production.

Risk ID	Risk category	Risk description	Potential impact	Mitigation strategy
R4	Privacy	Sensitive pricing or buyer procurement data may be exposed to unauthorized parties.	Loss of buyer trust and commercial risk.	Use channels, private data collections, RBAC, and careful API authorization checks.
R5	Operational	Physical quality verification may be disputed.	Payment settlement may be delayed by conflicts.	Integrate quality assessors, evidence upload, dispute workflow, and audit trail.
R6	Commercial	Buyers may be slow to shift from existing procurement relationships.	Reduced transaction volume and weak revenue generation.	Start with a focused niche pilot and demonstrate measurable ROI through traceability and payment reliability.
R7	Ethical	Reputation scores may disadvantage new farmers without historical data.	Unfair exclusion from market opportunities.	Apply initial neutral scores, explain scoring logic, and allow manual verification or probationary participation.
R8	Scalability	Prototype performance may degrade with large numbers of transactions.	Poor user experience and limited production readiness.	Benchmark transaction latency, optimize CouchDB indexes, scale peer nodes, and refine chaincode queries.